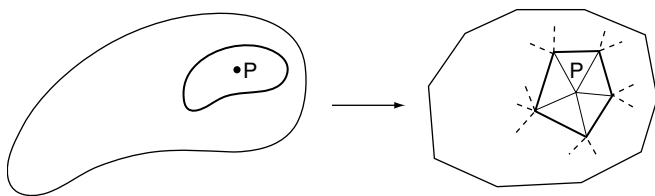


2

Closed surfaces

A surface is usually thought of as something smooth and rounded, like the surface of a sphere or a torus. In order to study surfaces systematically we shall, as described in the Introduction, assume that they are divided up into triangles.[†] It is therefore necessary at the outset to pinpoint the decisive property which an object must possess in order to be called a surface, and to interpret this as a property of the triangulation.

The decisive property is this. Take a point P on the surface; then the points of the surface close to P make up a little patch on the surface, this patch being essentially a two-dimensional disk. The disk may be bent or stretched, but must not be torn (that is, ‘essentially’ means, in technical language, ‘homeomorphic to’). In terms of triangulations this means that the edges opposite P in the triangles with P as a vertex form a simple closed polygon. A definition along



these lines is made precise in 2.1. Note that the definition excludes surfaces like the Möbius band and cylinder which have boundary edges or rims; to emphasize that our surfaces have no boundary we call them ‘closed surfaces’. The more general concept of ‘surface with boundary’ is explained and used in Chapter 9.

[†] This is no restriction for compact surfaces (e.g. surfaces which are closed subsets of a bounded region of euclidean space). Indeed with the more general concept of infinite triangulation a wider class of surfaces can be triangulated. These matters are dealt with in detail in Chapter I of the book by Ahlfors and Sario listed in the References and in the book *Elements of Combinatorial and Differential Topology* by Prasolov, also listed. There is also a proof in Doyle & Moran (1968).

In the formal definition of closed surface (2.1) the triangles will be ordinary flat triangles with straight edges. This is simply because flat triangles are easier to define and to work with than curvy ones. The reader should not feel uneasy because the smooth rounded surface has become a somewhat angular and kinky object; he is quite at liberty to continue drawing rounded surfaces – and indeed the author will exercise that liberty too. In Chapter 1 we saw that no generality was lost by specifying that the edges of a graph should be straight. In a similar way every surface with curved triangles can be realized (at any rate in $\mathbb{R}^5$) with flat triangles, but the question of realization is not taken up until 3.19.

The first objective of this chapter is to construct all closed surfaces and to exhibit them in a convenient way by means of a ‘polygonal representation’. From this representation it is possible, by a sequence of ‘scissors and glue’ operations to put every closed surface into exactly one of a list of standard forms – this list appears in 2.8. We can then go on to derive a geometrical description of closed surfaces in terms of three basic ingredients – sphere, torus and projective plane – and one mixing operation – connected sum. In this way we can picture what all closed surfaces ‘look like’.

Finally we show how to distinguish different closed surfaces from one another, and in particular how to prove that no two closed surfaces in the list 2.8 are the same.

There is an intrinsic interest in trying to describe all possible surfaces, but the longer-term purpose so far as this book is concerned is to provide plenty of good and geometrical examples with which to illustrate later theorems.

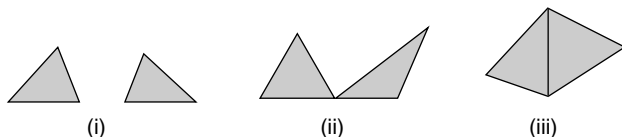
Closed surfaces and orientability

A graph in which there are no isolated vertices consists of a collection of closed segments (edges) which satisfy an intersection condition, namely that two edges intersect, if at all, in a common end-point (a vertex). We now introduce a two-dimensional building block and start to make two-dimensional objects, again restricting the possibilities by imposing an intersection condition. The new building blocks are triangles, which may be any shape or size. Note that any triangle comes equipped with three vertices and three straight edges. The *intersection condition* we impose is this:

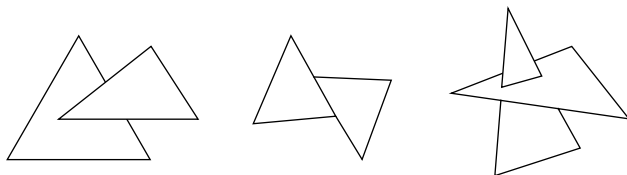
Two triangles either

- (i) are disjoint or
- (ii) have one vertex in common or

(iii) have two vertices, and consequently the entire edge joining them, in common.



Some configurations which are not allowed are pictured below:



(In the terminology of Chapter 3, a collection of triangles satisfying the intersection condition is said to form a pure two-dimensional simplicial complex.)

As in the case of graphs, we have the concept of connectedness: a collection M of triangles satisfying the intersection condition is called *connected* if there is a path along the edges of the triangles from any vertex to any other vertex. Indeed the set of edges and vertices of triangles in M forms a graph M^1 and we have just defined ' M is connected' to mean ' M^1 is connected'.

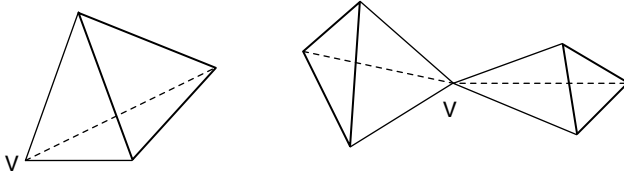
There is one other concept needed in order to define closed surfaces. Consider a vertex v of some triangle of a collection satisfying the intersection condition. The edges opposite v in the triangles of M having v as a vertex will form a graph, called the *link* of v .

Definition 2.1. A *closed surface* is a collection M of triangles (in some euclidean space) such that

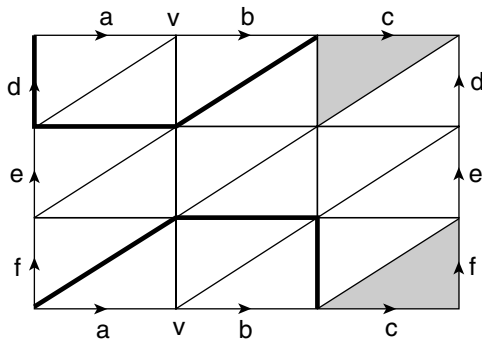
- (i) M satisfies the intersection condition (see above)
- (ii) M is connected
- (iii) for every vertex v of a triangle of M , the link of v is a simple closed polygon.

Remarks and Examples 2.2. (1) A hollow tetrahedron is a closed surface, the link of any vertex being the three edges of the opposite face. However two

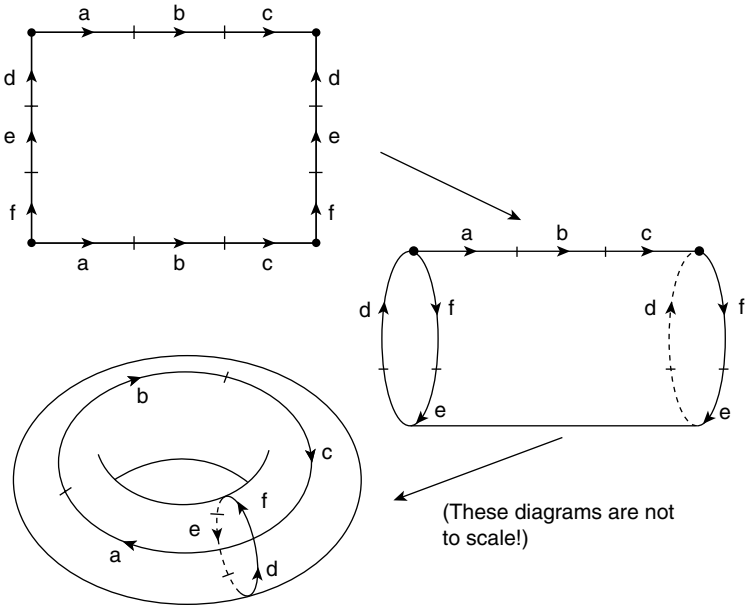
hollow tetrahedra with a common vertex do not form a closed surface since the link of the common vertex is two simple closed polygons and not one.



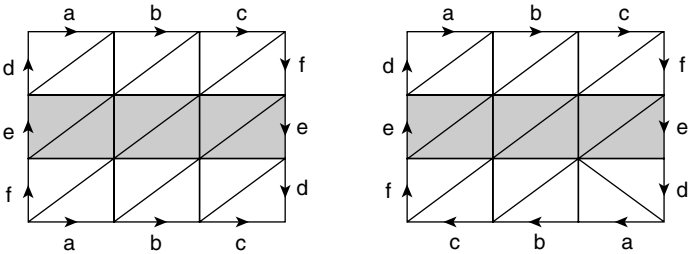
(2) It is not difficult to show that any edge of a closed surface M is on exactly two triangles of M . (Consider the link of one of the end-points of the edge.) Note that two hollow tetrahedra with a vertex in common have this latter property, but, as pointed out in (1), do not form a closed surface. Two hollow tetrahedra with an edge in common do not satisfy even the weaker condition that every edge is on exactly two triangles.

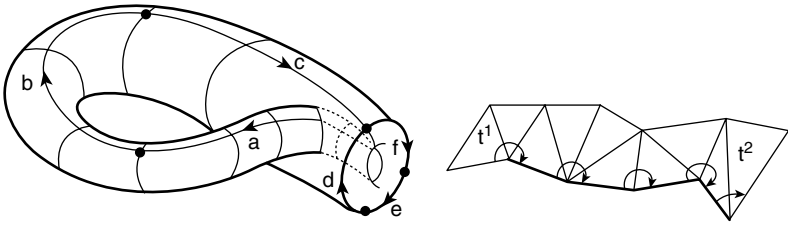


(3) The diagram above shows a schematic representation of a closed surface containing 18 triangles. The pairs of similarly marked edges are to be ‘identified’ with arrows corresponding so that, for example, the two shaded triangles actually have the edge c in common. The link of any vertex is then a simple closed polygon – the link of v is drawn heavily in the diagram, the two halves fitting together to form a simple closed hexagon. The closed surface so described is called a *triangulation of the torus* (sometimes abbreviated to ‘torus’) since, informally, if we stick together the similarly marked edges in $\mathbb{R}^3$, the result is a torus.



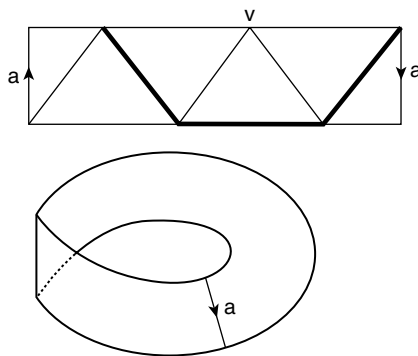
(4) The left-hand diagram below represents a closed surface which is a triangulation of the ‘Klein bottle’ and the right-hand diagram represents a triangulation of the ‘projective plane’. (The shading in these diagrams is not relevant until later.) Neither of these surfaces can be constructed in three-dimensional space (see 9.18). A picture of the Klein bottle, in which there is a self-intersection along a circle, appears on the next page. For similar pictures of the projective plane, see 2.7(3); see also the book of Hilbert & Cohn-Vossen listed in the References.





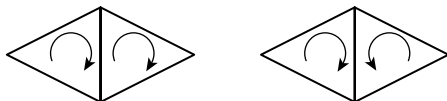
(5X) Let t^1 and t^2 be two triangles of a closed surface M . Join some vertex of t^1 to some vertex of t^2 by a simple path along the edges of M . Using the fact that the link of each vertex along this path is a simple closed polygon it is possible to construct a sequence of triangles connecting t^1 and t^2 , any two consecutive triangles in the sequence having an edge in common. The arrows in the right-hand diagram above show the way in which a sequence of triangles is constructed using in turn the links of vertices on the path. Note that such a sequence cannot always be constructed on the two tetrahedra with a common vertex, but it can on the two tetrahedra with a common edge.

(6) The definition of closed surface does not allow the collection of triangles in the upper diagram below, since the link of the vertex v is not a simple closed polygon (it is a polygonal arc). The diagram shows a triangulation of the *Möbius band*, drawn underneath, which is obtained by sticking together the ends of a rectangle after a twist of 180° . The Möbius band is a surface in a more general sense which will be explained in 9.8. Reversing one of the arrows in the upper diagram we obtain a *cylinder*, in which the ends of the rectangle are identified without a twist.



The major division amongst closed surfaces is into orientable and non-orientable closed surfaces. Once again we shall meet the concept of orientation more formally in the next chapter; for the present we shall use the following

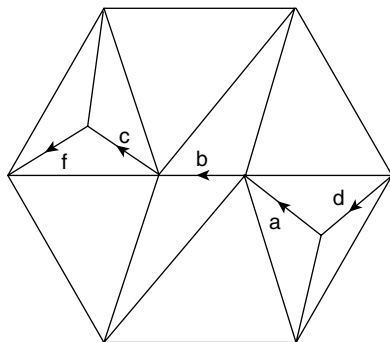
informal idea. Suppose that on each triangle of a closed surface M we draw a circular arrow (called an *orientation*) as in the diagrams.



In the left-hand diagram, the two triangles with a common edge are said to be *coherently oriented*, while in the right-hand diagram the two triangles are not coherently oriented.

Definition 2.3. A closed surface M is called (*coherently*) *orientable* if all the triangles of M can be given an orientation in such a way that two triangles with a common edge are always oriented coherently. Otherwise M is called *non-orientable*.

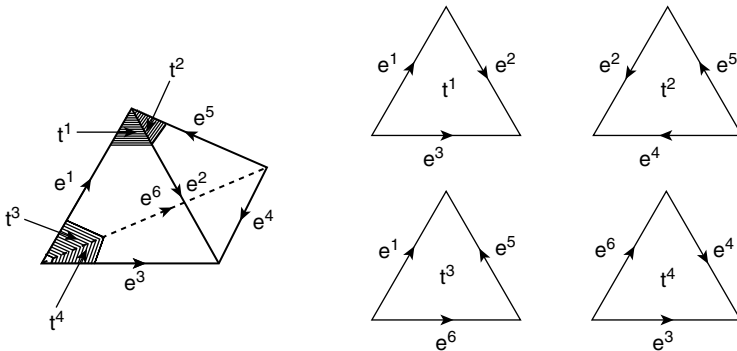
Example 2.4. (X) The torus in 2.2(3) is orientable, but the Klein bottle and projective plane in 2.2(4) are non-orientable. Note that the latter two closed surfaces both contain Möbius bands (see 2.2(6) and the shading in 2.2(4)). In fact it will become clear that a closed surface is non-orientable if and only if it contains a Möbius band. In the case of the Klein bottle, removing the shaded triangles in 2.2(4) leaves another Möbius band, as can be seen by identifying the two a 's, b 's and c 's. On the other hand the diagram below illustrates removal



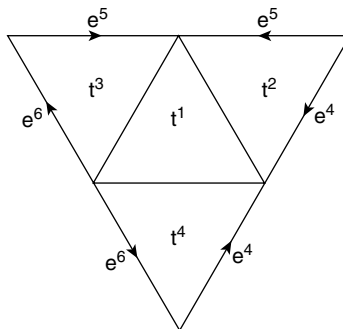
of the shaded triangles from the projective plane in 2.2(4) (allowing a little distortion of the triangles). This certainly does not contain any more Möbius bands. Thus the Klein bottle is 'more non-orientable' than the projective plane. In fact it is just twice as non-orientable, as we shall see following 2.9.

Polygonal representation of a closed surface

Let us start with a closed surface M , orient the edges arbitrarily and then label the triangles $t^1, \dots, t^n$ and their edges $e^1, \dots, e^m$ say. (In fact $3n = 2m$; in particular n is even.) The information needed to construct M comprises the n triangles and, for each triangle, the three oriented edges which belong to it. For example the tetrahedron on the left can be constructed from the four triangles on the right.

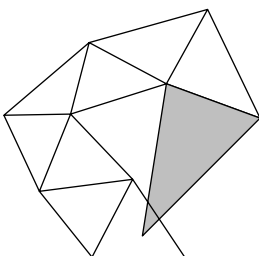


Our immediate object is to produce a more convenient plane representation of M than the collection of n disjoint triangles with labelled edges. This is achieved, in the case of the tetrahedron above, by partially assembling M in the plane: we could stick t^2 to t^1 along e^2 , then t^3 to t^1 along e^1 , then t^4 to t^1 along e^3 .



The result is a diagram which completely describes the way in which the triangles of M intersect one another. Other examples of such a plane representation are in 2.2(3), (4) and (6) above.

The same procedure can be followed for any surface M . Again we start with n disjoint triangles with labelled edges and assemble them one at a time. Triangles which have been assembled at a given stage will be called ‘used’ and the remainder ‘unused’. It is essential to identify at each stage an edge of an unused triangle to an edge of a used one; the boundary of the plane region covered by the used triangles will then always be a simple closed polygon. There is one slight snag: we may find that the unused triangle which is added at some stage overlaps the used triangles, as in the diagram, where the shaded



triangle was the last to be added. To overcome this we must abandon hope of having the triangles in the plane actually congruent to the triangles of M . Instead, as each triangle is used, we distort it (keeping the edges straight) so that the triangles in the plane do not overlap – for example by making the boundary polygon given by the used triangles always convex. The final representation of M will still faithfully record all the intersections of triangles of M (that is, it determines M ‘up to isomorphism’ in the language of Chapter 3). It remains to prove that, until all triangles have been used, there is always an unused triangle with an edge in common with the boundary polygon given by the used triangles. There is certainly an unused triangle with a vertex v in common with some used triangle, and the construction ensures that all vertices are on the boundary polygon. It is now straightforward to verify, using the fact that the link of v in M is a simple closed polygon, that there is an unused triangle with v and another vertex (and hence an edge) in common with a used triangle. (Note that the construction would break down at some stage if we started with two tetrahedra with a common vertex.)

Continuing until all the triangles of M are used, we obtain a *polygonal representation* of M as the region bounded by a simple closed polygon. This region is broken up into an even number n of triangles which correspond to the triangles of M and the boundary polygon has $n + 2$ edges which occur in equally labelled pairs. The identifications necessary to recover M can be described by a *symbol*: start at any vertex and read round the boundary polygon, writing an edge or

its ‘inverse’ according as its orientation goes with or against the direction of travel. A symbol for the tetrahedron above is

$$(e^5)^{-1}e^4(e^4)^{-1}(e^6)^{-1}e^6e^5.$$

Note that the torus in 2.2(3) could not arise directly from this construction since it has vertices not on the boundary polygon. What happens between the above construction and such a figure as 2.2(3) or 2.2(4) is that two edges with equal labels and orientation, and having a common vertex, become identified, creating an interior vertex. This process will form part of the reduction to standard form, below. The torus in 2.2(3) has a symbol $abcd^{-1}e^{-1}f^{-1}c^{-1}b^{-1}a^{-1}fed$, while the Klein bottle in 2.2(4) has a symbol $abcfedc^{-1}b^{-1}a^{-1}fed$. Note that a symbol is essentially cyclic: we could start reading it off at a different place; also read round the other way.

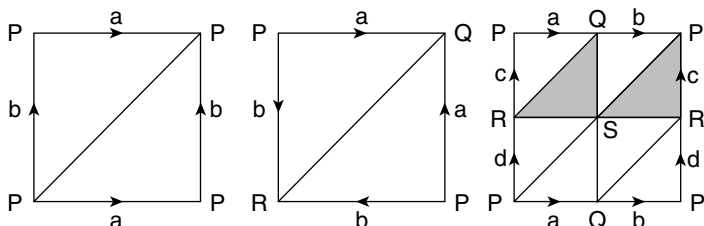
The triangles of a polygonal representation of a closed surface M can be oriented coherently (all clockwise, say, but the sceptical reader can prove it by induction on the number of triangles); whether this gives a coherent orientation of M itself depends on the disposition of arrows round the boundary polygon. If the symbol contains $\dots a \dots a^{-1} \dots$ then the orientation will be coherent across the edges making up a ; if it contains $\dots b \dots b \dots$ then the orientation will fail to be coherent across the edges making up b . In fact, we have the following result.

Proposition 2.5. *Suppose that a closed surface M is represented by a symbol, as above. Then M is orientable if and only if, for every letter occurring in the symbol, its inverse also occurs.* $\square$

★ A note on realizations

Suppose we are given a simple closed polygon with an even number of sides, the plane region bounded by the polygon being divided into triangles in such a way that the sides of the polygon are edges of triangles and two triangles intersect, if at all, in a vertex or an edge. Now mark the sides of the polygon with letters and arrows, each letter being used exactly twice as in the examples of 2.2(3), (4). The diagram can be interpreted as a ‘schema’ for sticking triangles together, the similarly marked edges being identified with arrows corresponding. When is it possible to realize this schema by a closed surface? The problem of realizing ‘schemas’ of this sort will be treated in more detail in Chapter 3 (see in particular 3.20); for the present we shall be content with stating the following result. The

identification of edges entails identification of certain vertices of the boundary polygon, and with this in mind we shall label the vertices of the diagram, those to be identified being given the same label, as in the diagrams below.



Note that none of these three determines a closed surface. For the left-hand and centre diagrams the two triangles would have all three edges in common, while for the right-hand diagram the two shaded triangles would have two vertices but no edge in common (see the intersection condition preceding 2.1).

Proposition 2.6. *A schema of the kind described above is realizable as a closed surface if and only if*

- (i) *the end-points of any edge are labelled differently, and*
- (ii) *no two edges (unless they are to be identified) have end-points labelled the same, and*
- (iii) *no two triangles have vertices labelled the same.* □

The diagram is then called a *polygonal representation* of the closed surface.

The polygonal representations described before are a special case of these; the identifications necessary to recover M can again be described by a symbol and the orientability criterion 2.5 still holds.

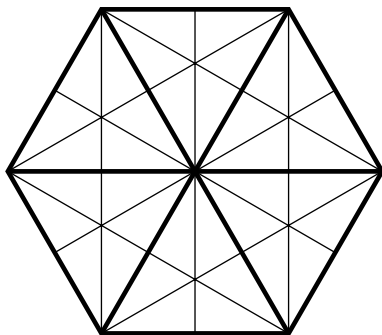
Note that (i)–(iii) imply that the link of any vertex will be a simple closed polygon. The reader may care to verify this, and also that the only case in which (i) and (ii) hold but (iii) does not is the centre diagram above.

It is worth noting that any diagram of the kind described before 2.6 can be subdivided into one which represents a closed surface: subdivide every triangle into six by drawing the medians, and then subdivide every small triangle into six again by drawing the medians. One such subdivision suffices for the centre and right-hand diagrams above, but both are needed for the left-hand diagram. The subdivision obtained by drawing the medians of all triangles is called *barycentric subdivision* and will be encountered again in Chapter 5. ★



Transformation of closed surfaces to standard form

We shall describe a method, traditionally known as ‘scissors and glue’, of reducing any polygonal representation of a surface to precisely one of a list of standard forms. During the course of this it may be necessary – or at any rate convenient – to subdivide M barycentrically (that is, drawing the medians of all triangles) a number of times; this is in order to ensure that our scissors always cut along edges of triangles of M . Thus the final symbol will represent either M or some repeated barycentric subdivision of M . Note that if M is a closed surface and M' is obtained from M by subdividing barycentrically then M' is still a closed surface. In the diagram, the link of the central vertex contains six edges of M (drawn heavily) and twelve edges of M' .

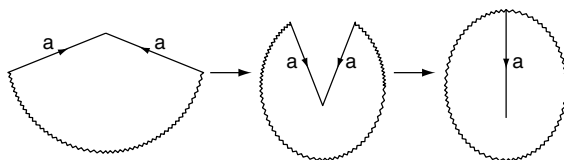


The reduction to standard form is in several steps, numbered as the parts of 2.7. The reader who turns straight to the answer (2.8) will not find himself much penalized in the sequel, since we shall not often make use of the precise nature of the steps used in the reduction to standard form; the main exception is the proof of 2.14.

2.7. The reduction algorithm

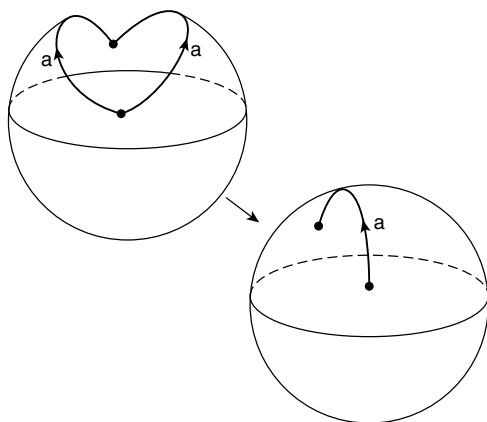
(1) First, a piece of notation. If a succession of oriented edges occurs twice on the boundary polygon, either in the same or opposite directions, then we shall always denote the succession of edges by a single letter. For example the diagram of 2.2(3) has symbol $xyx^{-1}y^{-1}$ where $x = abc$, $y = d^{-1}e^{-1}f^{-1}$. In what follows both x and x^{-1} (for example) are called *letters*. Note that, as in group theory, $(abc)^{-1} = c^{-1}b^{-1}a^{-1}$.

(2) There is a further analogy with group theory in that aa^{-1} can be ‘cancelled’ unless this would result in the symbol’s disappearing entirely. For the part aa^{-1} of a symbol containing at least four letters can be eliminated by ‘closing up’:

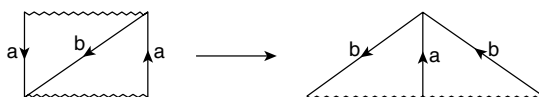


It may look as though this procedure will inevitably introduce curved edges into the diagram. In fact the resulting diagram will still be essentially a graph in the plane, and by Fáry's theorem (1.34) we can redraw it with straight edges and with the same triads of edges spanning triangles as before. In any case the diagram should really be treated as a schema for showing intersections of triangles, and it does not matter whether the edges are straight or curved.

The closing up operation is performed whenever possible during the succeeding stages. As for the symbol aa^{-1} itself, this is one of the final list of standard forms, and is to be regarded as a *sphere* (see diagrams).



(3) A pair $\dots a \dots a \dots$ is called a *pair of the first kind* and a pair $\dots a \dots a^{-1} \dots$ is called a *pair of the second kind*. A pair of the first kind can be brought together by the following rearrangement:



which replaces $\dots a \dots a \dots$ by $\dots bb \dots$. This process repeated produces

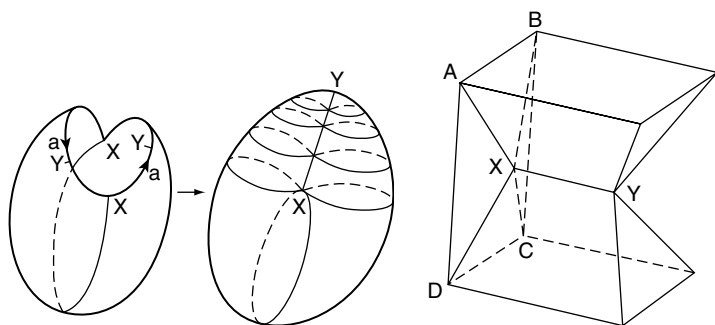
$$a_1 a_1 a_2 a_2 \dots a_p a_p X$$

where X consists entirely of pairs of the second kind, or is empty, and we say that $p = 0$ if there are no pairs of the first kind at all. The details of the repetition will be left to the reader, guided by the following hints:

$aab \dots b \dots$ can be turned into $aacc \dots$;

$aab \dots b^{-1} \dots$ can be turned into $aab \dots b \dots$ provided there is a pair of the first kind besides aa .

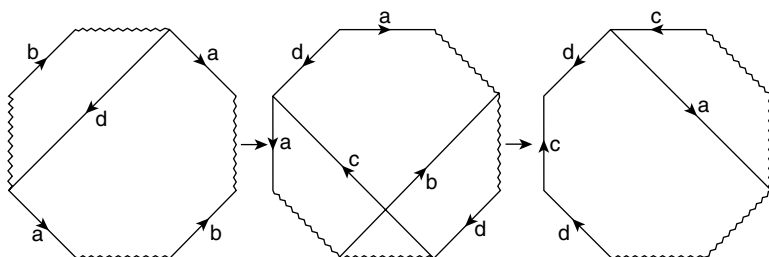
The symbol aa represents the closed surface obtained from a circular disk by identifying together each pair of diametrically opposite points on the boundary circle. This cannot be done in $\mathbb{R}^3$ (compare 2.16(5) and 9.18). A surface given by aa is known as a *projective plane* or *sphere with a cross-cap*. Any attempt to draw the surface in $\mathbb{R}^3$ will result in self-intersections, and two such attempts are pictured below. In each case the self-intersection is along the segment XY .



In both models, a is bent double, to lie along XYX . But the two halves, XY and YX , are on different sheets of the surface which intersect accidentally.

The right-hand surface consists of three rectangles (top, bottom and one end), four trapezia and four triangles (ABX and CDX are missing).

(4) If the symbol X in (3) above contains one pair of the second kind then either the whole symbol is aa^{-1} or, in view of (2), there is another pair of the second kind. Choosing $\dots a \dots a^{-1} \dots$ to be the pair which is closest together there must then be two 'interlocking' pairs $\dots a \dots b \dots a^{-1} \dots b^{-1} \dots$ (the b^{-1} could also come before a) and this can be turned into $\dots cdc^{-1}d^{-1} \dots$ by the following:

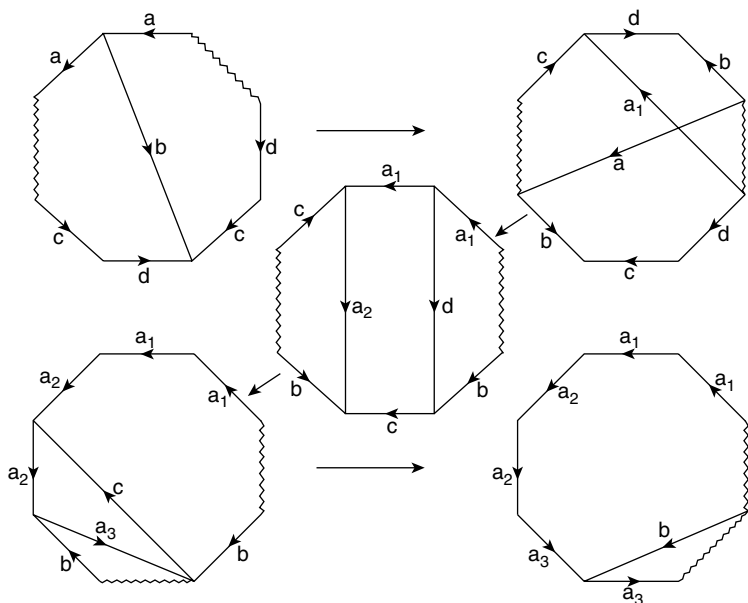


Note that no new pairs of the first kind can appear during this transformation, since no section of the figure, after cutting, is turned over before glueing back. In this way the symbol is turned into[†]

$$a_1 a_1 \dots a_p a_p c_1 d_1 c_1^{-1} d_1^{-1} \dots c_q d_q c_q^{-1} d_q^{-1}.$$

The symbol $cdc^{-1}d^{-1}$ represents a *torus* or *sphere with handle* (compare 2.2(3) where c appears as abc and d as fed , reading anticlockwise from bottom left).

(5) Finally, if there is at least one pair of the first kind then every $cdc^{-1}d^{-1}$ (handle) can be turned into two pairs of the first kind (cross caps):



Theorem 2.8. Every closed surface can (possibly after barycentric subdivisions) be represented by one of the following symbols:

- (1) aa^{-1} (sphere; orientable),
- (2) $a_1 a_1 a_2 a_2 \dots a_k a_k$, $k \geq 1$ (sphere with k cross-caps; non-orientable),
- (3) $a_1 b_1 a_1^{-1} b_1^{-1} a_2 b_2 a_2^{-1} b_2^{-1} \dots a_h b_h a_h^{-1} b_h^{-1}$, $h \geq 1$ (sphere with h handles; orientable). □

[†] The symbol given here has the property that the vertices which occur at the ends of the letters in the symbol all represent the same vertex of the closed surface. This comes about because at various points of the reduction pairs aa^{-1} are eliminated (compare (2) above), and a vertex which was on the boundary polygon becomes interior.

The number k or h is called the *genus* of the closed surface; the genus of a sphere is 0.

It is proved in 2.14 that any given closed surface can be reduced to only one of the forms given in 2.8.

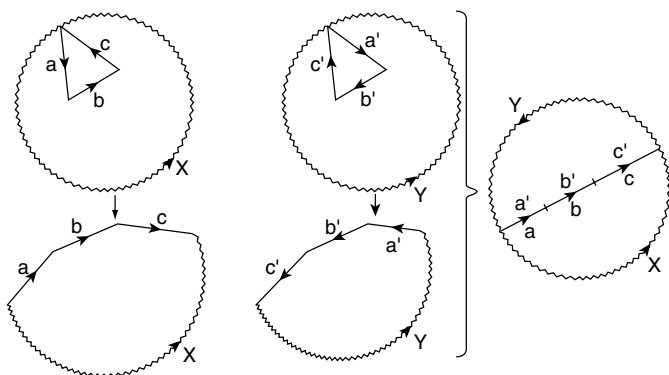
When a closed surface is represented by one of the above symbols we say that it is in *standard form*. Two closed surfaces M and N which reduce to the same standard form are called *equivalent*, and we write $M \approx N$. Of course two equivalent closed surfaces may be broken up into triangles in quite different ways. What $M \approx N$ says is that, forgetting about the ways in which M and N are broken into triangles, we can obtain each of them from a plane polygonal region with the same number of sides by making the same identifications, given by the symbols. Thus M and N ‘look the same’ – are in fact homeomorphic (see Chapter 5).

Any closed surface with standard form aa^{-1} is called a (triangulation of the) *sphere*; likewise closed surfaces with standard forms aa and $aba^{-1}b^{-1}$ are called (triangulations of) *projective plane* and *torus* respectively.

There is a simple geometrical interpretation for the juxtaposition of symbols, such as the succession of torus or ‘handle’ symbols $aba^{-1}b^{-1}$ which occurs in 2.8(3). Let M and N be closed surfaces given by symbols X and Y respectively (not necessarily in standard form). We shall construct a new surface with symbol XY obtained by juxtaposing X and Y . The new surface is far from being uniquely determined by M and N , but nevertheless its standard form is uniquely determined by those of M and N . This follows from 2.17 below.

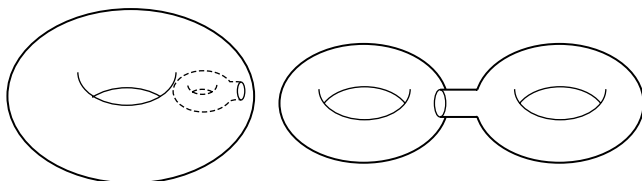
2.9. Connected sum of two surfaces

Choose a triangle of M meeting the boundary polygon of the plane representation with symbol X precisely in a vertex of M at the start of X



(if necessary barycentrically subdividing M), and similarly with N . Then remove these two triangles and glue together as in the diagram.

The result is a closed surface, orientable if and only if M and N are both orientable. It is called a *connected sum* of M and N . Informally we cut a hole in each surface and glue them together along the boundary rims. Thus for a connected sum of two tori there appear to be two choices:



In fact, as already claimed, these have the same standard form (indeed the little torus inside can be pulled through the hole to turn the left-hand picture into the right-hand one!). The surface is called a ‘double torus’, or ‘pretzel’. Thus the surface of 2.8(3) is an ‘ n -fold torus’ which looks like n tori joined by $n - 1$ tubes, and 2.8(2) is an ‘ n -fold projective plane’. In particular the Klein bottle, which has the usual symbol (compare 2.2(4)) $xyx^{-1}y$, has standard form $a_1a_1a_2a_2$, a ‘double projective plane’.

Let M and N be closed surfaces. The uniqueness property mentioned before 2.9 (and proved in 2.17) amounts to saying that any two connected sums of M and N are equivalent: we can speak up to equivalence of *the* connected sum of M and N .

Notation 2.10. We write $M \# N$ for any connected sum M and N .

Thus any two choices of $M \# N$ (for given M and N) are equivalent. Indeed, more is true: if M and N are each allowed to vary within an equivalence class then all the possible connected sums are still equivalent. This too is proved in 2.17.

The notation can be extended to several ‘summands’; again $M_1 \# \dots \# M_n$ is well-defined up to equivalence. We write nM for $M \# \dots \# M$ with n summands. Writing T for any closed surface with standard form $aba^{-1}b^{-1}$, i.e. for any torus, P for any projective plane and S for any sphere, the statement of 2.8 can be rephrased as follows:

every closed surface is equivalent to exactly one of S , kP or hT .

It can be shown that, for any closed surface M , we have $M \# S \approx M \approx S \# M$. This is most easily done using Euler characteristics (described below), but the reader can probably convince himself by drawing pictures. The same goes for the statement $P \# 2P \approx P \# T$ (this is essentially shown in 2.7(5)); note that

$2P \not\approx T$ so that ‘cancellation’ is not possible. Since $2P$ can be replaced by T provided this leaves at least one P , we can also rephrase 2.8 by saying that

every closed surface is equivalent to exactly one of

$S, hT, P \# \frac{k-1}{2}T$ (k odd), $K \# \frac{k-2}{2}T$ (k even, ≥ 2) where $K = 2P$ is a Klein bottle.

Euler characteristics

We shall prove that a given closed surface M cannot be reduced to two different standard forms 2.8. In addition to this we shall present a method by which the standard form of any surface given by a polygonal region with pairs of boundary edges identified (compare 2.6) can be found very quickly and without actually performing the scissors and glue operations of 2.7.

Definition 2.11. Let K be a collection of triangles satisfying the intersection condition given before 2.1. Writing $\alpha_0(K), \alpha_1(K), \alpha_2(K)$ for, respectively, the number of vertices, edges and triangles of K , the *Euler characteristic* of K is defined to be

$$\chi(K) = \alpha_0(K) - \alpha_1(K) + \alpha_2(K).$$

Proposition 2.12. *The Euler characteristic of K is unaffected by barycentric subdivision.*

Proof Write the α s after one subdivision as α' ; then

$$\alpha'_0 = \alpha_0 + \alpha_1 + \alpha_2, \quad \alpha'_1 = 2\alpha_1 + 6\alpha_2, \quad \alpha'_2 = 6\alpha_2.$$

Hence

$$\alpha'_0 - \alpha'_1 + \alpha'_2 = \alpha_0 - \alpha_1 + \alpha_2.$$

□

Our immediate aim is to calculate the Euler characteristic of the closed surface corresponding to any given symbol.

Suppose that a closed surface M has a symbol containing n letters (n is necessarily even), and is represented by a plane polygonal region bounded by a simple closed polygon with $n+r$ sides (each side of the polygon corresponding to an edge of M – recall that a letter in a symbol can stand for a succession of edges). The n vertices of the polygon at beginnings or ends of letters will represent some of the vertices of M ; let the number of distinct vertices of M amongst these be m .

Then

Theorem 2.13.

$$\chi(M) = m - \frac{1}{2}n + 1.$$

Remark. For the torus in 2.2(3), without any grouping into letters, $n = 12$, $r = 0$, $m = 5$. Grouping abc and fed into single letters $n = 4$, $r = 8$, $m = 1$ and in both cases $m - \frac{1}{2}n + 1 = 0$. The key fact is that once we have grouped edges into letters each vertex interior to a letter of M is identified only with the corresponding vertex interior to the other appearance of that letter.

Proof of 2.13 The set D of triangles assembled in the plane to give the plane representation of M gives rise to a graph which divides the plane into triangular regions. Write $\alpha_0(D)$, $\alpha_1(D)$, $\alpha_2(D)$ for the numbers of vertices and edges of this graph, and the number of triangular regions, respectively. (D stands for ‘disk’.) Euler’s formula (1.28) shows that $\alpha_0(D) - \alpha_1(D) + \alpha_2(D) = 1$ since there is one unbounded region. We write $\chi(D)$ for the left-hand side of this (compare 3.33).

The number of vertices of D on the boundary polygon is $n + r$ while the number of vertices of M on the boundary polygon is $m + \frac{1}{2}r$. Since vertices inside the polygon count equally in D and in M , it follows that $\alpha_0(M) - \alpha_0(D) = m - n - \frac{1}{2}r$. Similarly $\alpha_1(M) - \alpha_1(D) = -\frac{1}{2}n - \frac{1}{2}r$ and of course $\alpha_2(M) - \alpha_2(D) = 0$. Thus $\chi(M) - \chi(D) = m - \frac{1}{2}n$ and this, together with $\chi(D) = 1$, gives the result. $\square$

Corollary 2.14. *The Euler characteristic of any closed surface with standard form*

$$2.8 (1) \quad \text{is } 2;$$

$$2.8 (2) \quad \text{is } 2 - k;$$

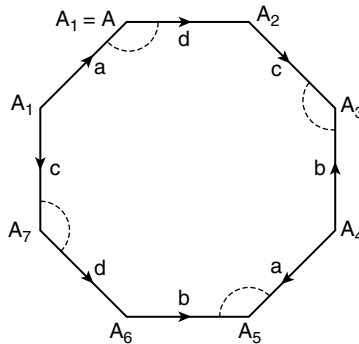
$$2.8 (3) \quad \text{is } 2 - 2h.$$

The standard form of any closed surface M is unique, and is determined by $\chi(M)$ and the orientability or otherwise of M .

Proof For the first statement, it is enough to calculate the Euler characteristics of the closed surfaces given by 2.8, since (as the reader may verify) the scissors and glue operations of 2.7, as well as barycentric subdivision, do not affect the Euler characteristic. In case (1), $m = 2$, while in cases (2) and (3), $m = 1$. The result now follows from the theorem.

For the second statement, note that the scissors and glue of 2.7 does not affect orientability either. Since no two standard forms agree in point of orientability and Euler characteristic, the result follows. $\square$

Remark 2.15. When calculating the number m of 2.13 the procedure of the following example is used. Choose any vertex (at beginning or end of a letter) and call it A , say $A_1 = A$.



A_1 is at the end of a ; so is A_5 . Hence in M , $A_5 = A$.

A_5 is at the end of b ; so is A_3 . Hence in M , $A_3 = A$.

A_3 is at the end of c ; so is A_7 . Hence in M , $A_7 = A$.

A_7 is at the beginning of d ; so is A_1 : we are back where we started.

The important thing is that no other vertex can be A , since in M the link of A is one simple closed polygon, which is indicated by the dotted lines in the diagram. How start at another vertex, say $A_2 = B$ and deduce $A_4 = A_6 = A_8 = B$. Hence $m = 2$, so that $\chi(M) = -1$ and since M is non-orientable (it contains $\dots b \dots b \dots$), M has standard form $3P$, i.e. $a_1 a_1 a_2 a_2 a_3 a_3$. Thus M is a sphere with three cross-caps.

Examples 2.16. Check the following, (1)–(4).

(1X) $abcbca$ is a Klein bottle, i.e. has standard form $2P$.

(2X) $abca^{-1}cb^{-1}$ has standard form $3P$.

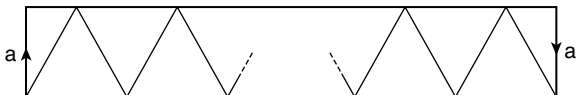
(3X) $abcdefe^{-1}db^{-1}afc$ has standard form $6P$.

(4X) $ae^{-1}a^{-1}bdb^{-1}ced^{-1}c^{-1}$ has standard form $2T$.

★(5X) Since any orientable closed surface is equivalent to a connected sum of tori (see the discussion following 2.10) it is clear that a representative of each equivalence class can be realized in $\mathbb{R}^3$. In Chapter 9 (see 9.18) we shall sketch a proof of the theorem that no non-orientable closed surface can be realized in $\mathbb{R}^3$. However they can all be realized in $\mathbb{R}^4$, as follows. Since every non-orientable closed surface is equivalent to a connected sum of tori and either a

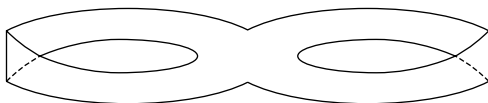
protective plane P or a Klein bottle K it is enough to check that P and K can be realized in $\mathbb{R}^4$,

For P , start with a Möbius band, triangulated as in the diagram.



It is clear that with enough triangles this can be realized in $\mathbb{R}^3$ (can the reader discover the minimum number needed?). Now take coordinates (x, y, z, t) in $\mathbb{R}^4$ and let the Möbius band be realized in the space $t = 0$. We add triangles as follows: for each pair v, w of vertices consecutive along the rim of the band, add the triangle with vertices v, w and $(0, 0, 0, 1)$. The result is a closed surface (for example the link of $(0, 0, 0, 1)$ is the rim of the band, which is a simple closed polygon) and the Euler characteristic is easily verified to be 1. Hence it is a projective plane. We have ‘added a cone’ to the boundary rim of the band. Cones will be used on several occasions later in the book; see the index for references.

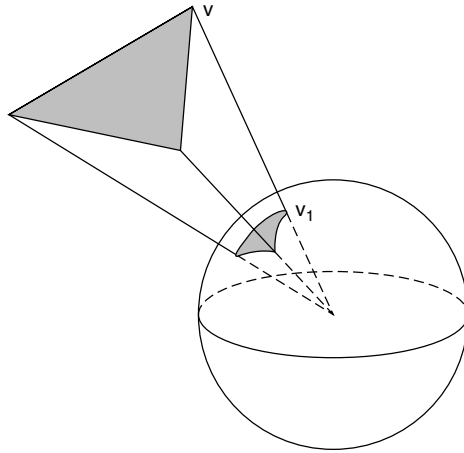
The Klein bottle K can be constructed in $\mathbb{R}^4$ in a similar way, starting with a triangulation of the ‘double Möbius band’ – two bands with parts of their rims identified. The boundary rim of the double band is still a single closed curve, and adding a cone gives a non-orientable closed surface of Euler characteristic 0, hence a Klein bottle.



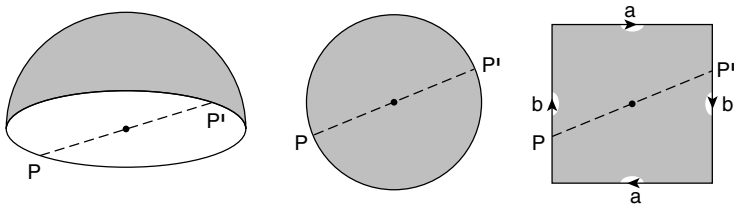
An alternative approach is to start with the picture of a Klein bottle in 2.2(4), where there is self-intersection along a circle. Using the fourth spatial dimension one of the intersecting tubes can be ‘moved aside’ to miss the other. (This is analogous to taking two intersecting lines in a horizontal plane and moving one of them slightly upwards to miss the other.) ★

★(6X) Let M be a closed surface in $\mathbb{R}^3$, not passing through the origin 0 , and with the property that every half-line (ray) through 0 meets M precisely once. Then M is a triangulation of the sphere, i.e. has standard form aa^{-1} . One way to see this is as follows. For each vertex v of M let v_1 be the (unique) intersection of the half-line $0v$ with a fixed sphere centred at 0 . Thus $v \rightarrow v_1$ projects the vertices of M on to the sphere. Similarly the edges of M project to arcs of great circles on the sphere and triangles to spherical triangles. Since half-lines meet M exactly once we obtain a collection of (spherical) triangles covering the sphere, with intersections exactly as in M . To recover the standard

form aa^{-1} we can remove one spherical triangle t and flatten separately t , and the rest of the sphere, on to a plane (e.g. by stereographic projection). This gives a plane representation of M by t and a triangular region which is itself broken up into triangles. Re-identifying one edge of t with one boundary edge of the triangular region gives the standard form aa^{-1} . It is possible to avoid curved triangles in this argument, though of course in the plane they can in any case be straightened by Fáry's theorem (1.34). ★



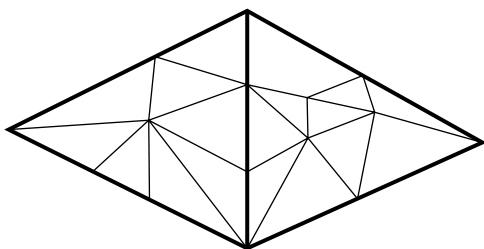
★(7X) Suppose that the closed surface M_1 in (6) has the property that it is 'symmetric about 0', i.e. that it is left unchanged by reflexion through the point 0. Then, at any rate provided M_1 has enough triangles, we can form a closed surface M_2 by identifying opposite triangles of M_1 . Since $\chi(M_1) = 2$ it is clear that $\chi(M_2) = 1$, so that M_2 must be a triangulation of the projective plane. To see this informally, think of identifying diametrically opposite points all over the sphere. We can throw away the southern hemisphere provided we leave the equator and identify opposite points of it. Flattening out the hemisphere we get a disk, or a square, with opposite points of its boundary identified. The latter is



Each diametrically opposite pair P, P' is identified to a point

one of the usual pictures of a projective plane (2.2(4)). ★

★(8XX) Let M be a closed surface. We form a *subdivision* M_1 of M , by dividing up every triangle of M into smaller triangles. We require the subdivisions of any two adjacent triangles to be consistent on their common edge as in the diagram. A special subdivision is the barycentric subdivision used in this chapter. (See also Chapter 5.) Let $\bar{\alpha}_0(M_1), \bar{\alpha}_1(M_1)$ denote the numbers of vertices and edges of M_1 lying in edges of M . Show that $\bar{\alpha}_0(M_1) - \bar{\alpha}_1(M_1) = \alpha_0(M) - \alpha_1(M)$. Note that, if D denotes the part of M_1 lying within a single triangle of M , then $\chi(D) = 1$; compare the proof of 2.13. Deduce that $\chi(M) = \chi(M_1)$: this says that the Euler characteristic is invariant under subdivision. ★



Now consider the effect of making a connected sum (2.9) on Euler characteristics. Since barycentric subdivision does not affect Euler characteristics (2.12) it is immediate from the definition that $\chi(M_1 \# M_2) = \chi(M_1) + \chi(M_2) - 2$ for any connected sum $M_1 \# M_2$ of two closed surfaces M_1 and M_2 . It is also true that $M_1 \# M_2$ is orientable if and only if M_1 and M_2 are both orientable: thus orientability and Euler characteristic for $M_1 \# M_2$ are determined by M_1 and M_2 . By 2.14 we have:

Proposition 2.17.

- (1) Any two connected sums of M_1 and M_2 are equivalent.
- (2) If $M_1 \approx N_1$, and $M_2 \approx N_2$ then

$$M_1 \# M_2 \approx N_1 \# N_2.$$

□

★ Minimal triangulations

Among all triangulations of, say, a torus – that is, among all closed surfaces M with standard form $aba^{-1}b^{-1}$ – there will be some which have the fewest possible triangles. In this section we investigate such ‘minimal triangulations’ briefly and relate them to a famous problem of graph theory.

Let us fix attention on a particular standard form, which has Euler characteristic χ given by 2.14. Any closed surface M with this standard form has the following properties:

$$2.18 \quad \begin{cases} \alpha_1(M) \leq \frac{1}{2}\alpha_0(M)(\alpha_0(M) - 1) & \text{(see the note following 1.1)} \\ 3\alpha_2(M) = 2\alpha_1(M) & (2.2 \text{ (2)}) \\ \alpha_0(M) - \alpha_1(M) + \alpha_2(M) = \chi & (2.11) \end{cases}$$

These give $\alpha_0^2 - 7\alpha_0 + 6\chi \geq 0$. With the exceptions noted below, this is equivalent to

$$2.19 \quad \alpha_0 \geq \frac{1}{2}(7 + \sqrt{49 - 24\chi}).$$

The exceptions are as follows:

If $\chi = 1$ there is a solution $(\alpha_0, \alpha_1, \alpha_2) = (1, 0, 0)$

If $\chi = 2$ there are solutions $(\alpha_0, \alpha_1, \alpha_2) = (2, 0, 0)$ and $(3, 3, 2)$.

However none of these exceptions corresponds to a genuine surface. The only one that makes sense at all is $(3, 3, 2)$ for the sphere, and this is two distinct triangles with the same three vertices and the same three edges, a configuration which cannot be realized with flat triangles and which violates the intersection condition given before 2.1.

Note that $\alpha_0(M)$ is minimal if and only if $\alpha_2(M)$ is minimal, and also that equality occurs in 2.19 if and only if it occurs in the first relation of 2.18.

Examples 2.20. (1) Consider the standard form aa^{-1} : the sphere. Then $\chi = 2$ and from 2.19, $\alpha_0(M) \geq 4$. In fact there is a triangulation, namely the hollow tetrahedron, with exactly four vertices.

(2) Consider the standard form aa : the projective plane. Then $\chi = 1$ and $\alpha_0(M) \geq 6$. A triangulation with six vertices is drawn in 4.17(5).

(3) Consider the standard form $aba^{-1}b^{-1}$: the torus. Then $\chi = 0$ and $\alpha_0(M) \geq 7$. A triangulation with seven vertices is drawn in 3.22(1). There is another standard form with $\chi = 0$, namely $a_1a_1a_2a_2$: the Klein bottle. Is there a triangulation with seven vertices? In fact not: the smallest possible number is eight, and it is not hard to find triangulations with eight vertices. This is a curiously exceptional example, for in every other case where the right-hand side of 2.19 is an integer there does exist a triangulation with exactly that number of vertices. Not surprisingly, this is far from easy to prove; we shall shortly sketch a proof which depends on a very difficult theorem in graph theory, and also on a result to be proved in Chapter 9.

(4X) Show from 2.19 that

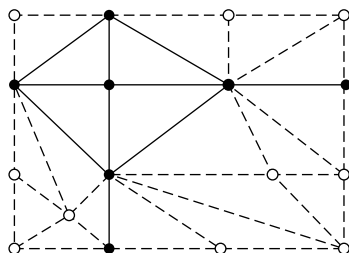
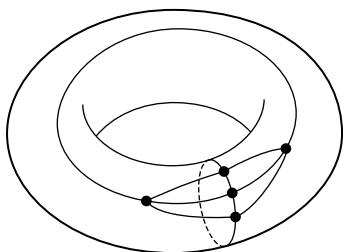
$$\alpha_0(M) \geq \left\lfloor \frac{1}{2} (9 + \sqrt{48 - 24\chi}) \right\rfloor$$

where $\lfloor x \rfloor$ means the greatest integer $\leq x$.

(5) In the non-orientable case, Ringel (1955) has determined precisely the number of vertices in a minimal triangulation. It is $\left\lfloor \frac{1}{2} (9 + \sqrt{48 - 24\chi}) \right\rfloor$ for $\chi \neq 0, -1$; and 8 for $\chi = 0$ (Klein Bottle); 9 for $\chi = -1$. For the orientable case Jungerman & Ringel (1980) establish a formula for the number of vertices in a minimal triangulation, namely $\left\lceil \frac{1}{2} (7 + \sqrt{49 - 24\chi}) \right\rceil$ where $\lceil \cdot \rceil$ is the ‘ceiling’ function taking a real number x to the smallest integer $\geq x$. There is an exception here too: for $\chi = -2$ the number is 10.

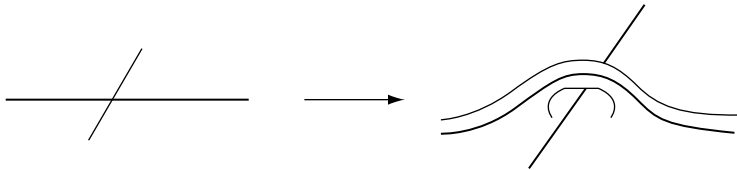
Given a graph G we can enquire, for example, ‘Is it possible to embed G in the torus?’ This means ‘Is there some triangulation M of a torus with the property that the graph consisting of the set of vertices and edges of M contains a subgraph isomorphic to G ?’ Thus for example let G_n denote the complete graph with n vertices (every pair of vertices being joined by an edge); then for $n \leq 7$, G_n can be embedded in a torus, for the triangulation drawn in 3.22(1) has the property that its set of vertices and edges is a G_7 . Indeed, whenever we have a triangulation for which equality holds in 2.19 it will satisfy $\alpha_1 = \frac{1}{2}\alpha_0(\alpha_0 - 1)$, and therefore the set of vertices and edges of the triangulation will be a complete graph G_{α_0} .

There is an alternative, less formal, way in which we can think of embedding a graph in a closed surface. We can think of the surface in its usual rounded shape and the graph drawn with curved edges in the surface. For example the left-hand diagram below shows G_5 in the torus. In fact, the two ideas are equivalent: a graph can be embedded in this way in a curved surface if and only if it can be embedded in the way previously defined as a straight graph contained in some triangulation. Roughly speaking, given a graph in a curved surface, we



add extra vertices and edges away from the graph until the surface is divided up into curved triangles, and then regard this as a schema for the construction of a genuine (straight) triangulation (see 2.6). This is analogous to the construction of a triangular graph in the plane (1.34(I)) except for the introduction of extra vertices. One way of carrying this out for G_5 in the torus is illustrated above.

In the spirit of the preceding paragraph, let us see how to embed an arbitrary graph in a closed surface of sufficiently high genus (2.8). Given a graph G in $\mathbb{R}^3$ (see 1.4), project G from some point in $\mathbb{R}^3$ on to a plane, choosing the point of projection O in such a way that the ray through O and a vertex of G does not meet G again, while the ray through O and any other point of G meets G again at most once. For example, starting with G_5 we might obtain the diagram in 1.5(1) in the plane. At each crossing point (not the projection of a vertex) we stick a small handle on the plane:



In this way all the crossings are eliminated, and enclosing the projected graph in a large circle which is then capped by a hemisphere we obtain a graph isomorphic to G realized in a sphere with handles (or with crosscaps, if a crosscap is added somewhere). Applying this procedure to 1.5(1) we can embed G_5 in an orientable surface of genus 5 – far from the best result, for it also embeds in the torus, which has genus 1.

We are interested, in what follows, not in whether a graph embeds in a particular surface, but in whether it embeds in some surface in a given equivalence class – the notion of equivalence being that given after 2.8 (i.e. having the same standard form). Thus unless otherwise stated we regard all surfaces as variable within a definite equivalence class. This most accurately reflects the informal idea of drawing the graph on a curved surface.

It is not an easy matter to determine the surface of smallest genus in which a given graph can be embedded. A remarkable theorem, due essentially to G. Ringel and J. W. T. Youngs, but in which several other people had a hand, determines just this for the complete graph G_n . (See Ringel & Youngs (1968), Ringel (2002) and the books of Ringel, Bollobás and Gross & Yellen listed in the References.) The following statement of the theorem is directed at the opposite problem: given a surface (variable of course within an equivalence class) what is the largest complete graph which can be embedded in it?

Theorem 2.21 (Ringel–Youngs). *The largest complete graph which will embed in a closed surface, other than the Klein bottle, of Euler characteristic χ , is the one with*

$$\mathcal{H}(\chi) = \left\lfloor \frac{1}{2} (7 + \sqrt{49 - 24\chi}) \right\rfloor$$

vertices (as usual $\lfloor \cdot \rfloor$ stands for the integer part). In the case of the Klein bottle, the largest complete graph is G_6 , not G_7 . $\square$

Note the anomalous position of the Klein bottle, and also the similarity between 2.21 and 2.19. I claim, in fact, the following.

Theorem 2.22. [†] *Suppose that $\frac{1}{2}(7 + \sqrt{49 - 24\chi})$ is an integer, λ say, where χ is an integer ≤ 2 . Then there is a closed surface with λ vertices and Euler characteristic χ . Indeed if $\chi < 0$ is even then there are two such, one orientable and one non-orientable. If $\chi = 0$ then there is only one, which is a torus. The set of vertices and edges of the surface is a complete graph G_λ .*

Note that here the claim is that G_λ embeds in a particular closed surface, not merely one of a given equivalence class. Note also that the triangulation is minimal by 2.19. In fact the theorem above asserts that the lower bound of 2.19 is attained provided $\frac{1}{2}(7 + \sqrt{49 - 24\chi})$ is an integer.

Here is a sketch of a proof of 2.22. It follows from the Ringel–Youngs theorem that, with $\chi \neq 0$, G_λ embeds in any closed surface with characteristic χ but not in any with characteristic $> \chi$. The case $\chi = 0$ requires special attention. From this it can be shown that whenever G_λ is embedded in a closed surface of characteristic χ , the graph will divide it up into triangles: that is, G_λ forms the vertices and edges of triangulation in which the triangles are the triangular regions into which G_λ divides the surface. A proof of this is sketched in the last paragraph but one of 9.28(7), which depends on several results of Chapter 9. Using the fact that G_λ is a graph there is no difficulty in checking that the triangles satisfy the intersection condition given before 2.1. (The only possible exception is G_3 , embedded in the sphere to give two (curved) triangles meeting in all three edges, but fortunately $\lambda = 3$ is not a possible value. This anomalous case turned up in the discussion of 2.19 as well.) $\square$

Examples 2.23. (1) Take $\chi = -10$ in 2.22. Then $\lambda = 12$; hence there are surfaces equivalent to $6T$ and to $12P$ (both of which have characteristic -10)

[†] My aim in this brief exposition is to display some connexions between the Ringel–Youngs theorem, minimal triangulations and (shortly) colouring problems. I hasten to point out that existing proofs of the Ringel–Youngs theorem proceed on a case by case basis, and that one case is close in content to the present theorem.

having 12 vertices. In each case the set of vertices and edges is G_{12} : this result says that starting with G_{12} we can build up both an orientable and a non-orientable closed surface by filling in suitable triangles. (The G_{12} had better be in a high-dimensional space to avoid accidental intersections of triangles. In fact a theorem in Chapter 3 (3.19) says that five dimensions will do.)

(2X) For which values of χ is $\frac{1}{2}(7 + \sqrt{(49 - 24\chi)})$ an integer? (Answer: $\chi = (7n - n^2)/6$ where $n \equiv 0, 1, 3$ or $4 \pmod{6}$.)

(3X) Use 2.21 to find explicitly the Euler characteristics of the orientable and non-orientable surfaces of lowest genus into which it is possible to embed G_n .

It is worth pointing out the connexion between 2.21 and the *colouring problem* for graphs in a closed surface. A graph is n -colourable if each vertex can be assigned one of n colours in such a way that two vertices at the ends of an edge are always coloured differently. Thus for example a graph with n vertices is certainly n -colourable, and a complete graph G_n is not m -colourable for any $m < n$. Now consider all graphs which can be embedded in a given closed surface (or, as usual, in any equivalent one). What is the smallest n for which all these graphs are n -colourable? This number is called the *chromatic number* of the closed surface. Clearly, from 2.21 the chromatic number of any closed surface of Euler characteristic χ is at least $\mathcal{H}(\chi)$ (except for the Klein bottle, when it is at least 6.) In fact Heawood (1890) proved that the chromatic number is $\leq \mathcal{H}(\chi)$, for any closed surface except the sphere. (See 9.28(8).) Heawood actually believed, incorrectly, that he had proved the chromatic number equal to $\mathcal{H}(\chi)$, and the equality became known as *Heawood's conjecture*. Franklin (1934) found a counterexample, when he showed that the chromatic number of the Klein bottle is 6, not $\mathcal{H}(0) = 7$. The Ringel–Youngs theorem showed, astonishingly, that Franklin's counterexample was the only one. Thus the chromatic number is

$$\begin{array}{ll} \mathcal{H}(\chi) & \text{for any closed surface except Klein bottle or sphere} \\ 6 & \text{for the Klein bottle} \\ \geq 4 & \text{for the sphere.} \end{array}$$

The sphere is not covered by Heawood's theorem: in fact in his 1890 paper Heawood showed only that the chromatic number of a sphere is ≤ 5 . It had been conjectured around 1852 that the chromatic number was 4 and indeed another function of Heawood's paper was to expose a fallacious proof of this conjecture. For over 100 years the notorious 'four colour conjecture' remained unsolved. A proof was found in June 1976 and published the following year, but even then the proof used high speed computers and it is a matter for speculation

whether a purely conceptual proof that can be checked by a human being will ever be found. At any rate the *four colour theorem* asserts that the chromatic number of the sphere is 4. This is the same as for graphs in the plane.

For an account of the proof by the original authors see Appel & Haken (1977); see also the book *Four Colours Suffice* by Wilson listed in the References. For the five colour theorem see for example the book *Introduction to Graph Theory* by Wilson. For the history of the four colour conjecture, and the connexion with geographical maps, see the book of Biggs, Lloyd & Wilson.

